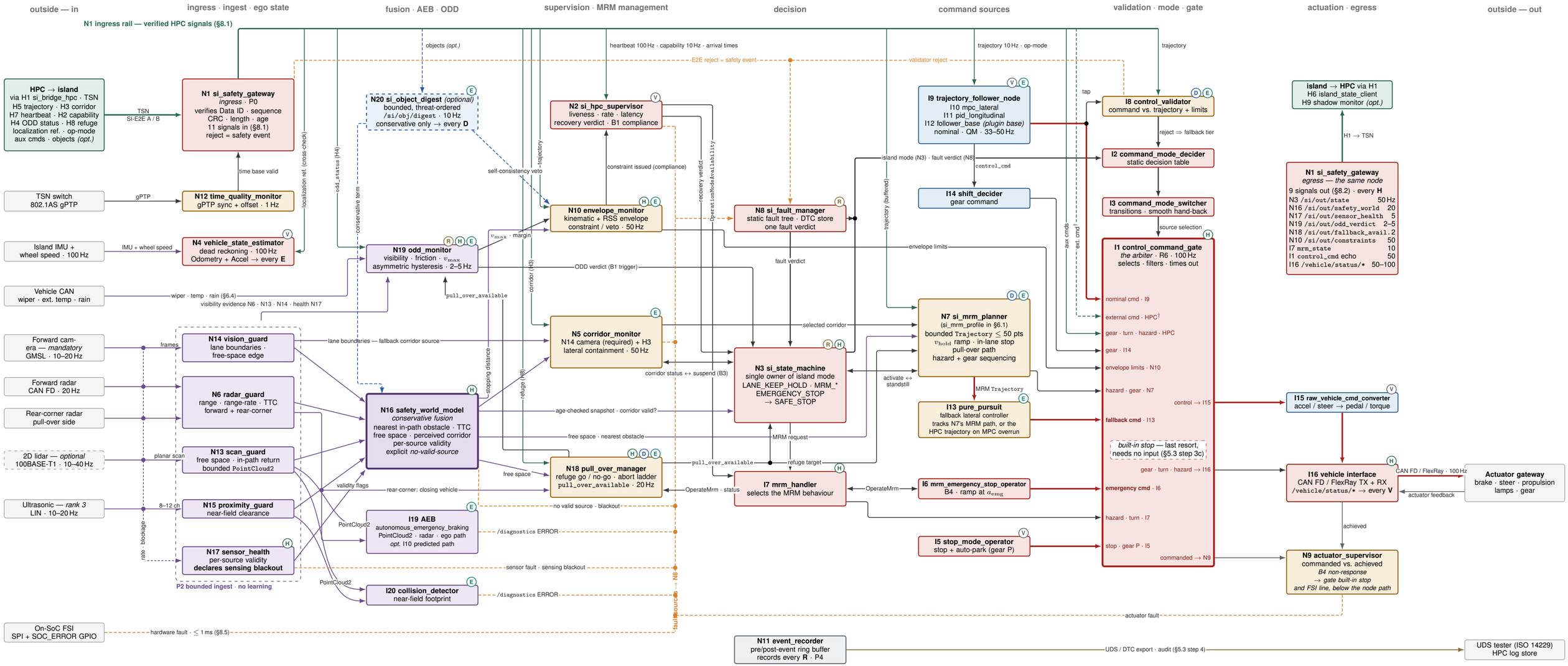
L4 Safety Island — message flow across every island node

Extends §10.2 *The island sensing chain* from the sensing nodes to all 37 island nodes — 18 migrated Autoware **I**-nodes and 19 new **N**-nodes — plus the optional N20.
Drawn from `L4SafetyIsland_SoftwareDesign.md` (version 20261023) without modifying it. Every edge, with the section it is taken from, is catalogued in `L4SafetyIsland-MessageFlow.md`.
Companion — `../nodediagram/` redraws Autoware's own node diagram with the island added: *where* each node runs and what changed, coloured by Autoware stack. **Here, colour is the §9.1 execution partition and every message is drawn.**



Nodes (partition, §9.1)

P0 safety core ASIL-D · 100 Hz | P1 supervision + fallback · ASIL-B | P2 safety sensing ASIL-B · 20–50 Hz | P2 fusion point N16 | P3 nominal QM, monitored | P4 non-real-time QM | optional (§3.5) may be absent | outside the island

Messages

- island sensor data
- HPC-sourced, verified by N1
- control command path
- optional, conservative-only
- state, mode, decision
- request ↔ status pair
- fault / `/diagnostics` → N8
- actuator feedback

Connectors (a badge replaces a line)

- (E) subscribes to N4 island ego state — `Odometry + AccelWithCovarianceStamped`
- (V) subscribes to I16 `/vehicle/status/{velocity,steering,gear,control_mode}_status`
- (D) may take the optional N20 digest — narrows only, never a precondition
- (H) published to the HPC through N1 egress (list in the N1 egress box)
- (R) recorded by N11 — DTCs, state transitions, SOTIF disagreement

† The external / remote command reaches the gate from the HPC (§5.2, §10.4) but has no row in the §8.1 interface table. No topics of their own: I4 command_mode_types (a library linked into I1–I3), I10–I12 (plugins inside I9). Legacy path: I17 + I18 replace I1–I3 and are not drawn.
Not drawn here, because the companion draws it: the HPC side of the link (H1–H9), the Autoware stacks that stay on the HPC, and what changed against Autoware as it is today — `../nodediagram/`. Use that one for the allocation argument, this one for timing and interfaces.